

Integrated Analysis and Simulation of Buck-Boost Converter Performance

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INDEX

- I. Cover Page**
- II. Abstract**
- III. Introduction**
- IV. Literature Review**
- V. MOSFET Characteristics Analysis**
- VI. Analysis of Duty cycle for different voltage configurations**
- VII. Simulation of a Buck-Boost Converter Circuit**
- VIII. Analysis and MATLAB Visualization of Converter Ratio and Efficiency in Buck-Boost Converters**
- IX. Design and simulation results**
- X. Derivation of Buck boost converter**
- XI. Conclusion**

ABSTRACT

The Buck-Boost converter is an essential type of DC-DC converter used in power electronics for efficiently adjusting and controlling voltage levels. Unlike standard converters like the boost converter, which only increases voltage, or the buck converter, which only decreases it, the buck-boost converter has the capability to both step up and step down voltage. This makes it extremely valuable across various manufacturing applications. This report thoroughly examines the key components of buck-boost converters, including their topology and design considerations. It also explores the characteristics of MOSFETs using simulation circuits and output waveforms to highlight important performance and efficiency aspects. With detailed insights into the design and operational features of buck-boost converters, this report seeks to provide engineers and researchers with the necessary knowledge to effectively utilize these converters in power electronics applications.

Type of MOSFET Used:

For the implementation of the Buck-Boost converter, the IRFZ44N MOSFET is selected. The IRFZ44N is a power MOSFET with low on-resistance and high switching speed, making it suitable for high-efficiency switching applications like power converters. Its robust design and high current-carrying capability ensure reliable operation in the Buck-Boost converter circuit.

About LTspice:

LTspice, a powerful simulation tool developed by Linear Technology Corporation, is utilized for modeling and simulating the Buck-Boost converter circuit. LTspice offers a user-friendly interface and comprehensive library of electronic components, enabling accurate simulation of complex circuits. Its transient analysis capabilities allow for dynamic evaluation of circuit performance under varying operating conditions, providing valuable insights into efficiency, voltage ripple, and control mechanisms of the Buck-Boost converter. With its extensive features and capabilities, LTspice serves as an indispensable tool for designing and optimizing power electronics circuits for renewable energy applications like solar energy conversion.

Introduction

The global transition towards renewable energy sources, driven by concerns over climate change and energy security, has accelerated the demand for efficient energy conversion technologies in solar applications. Solar energy, abundant and inexhaustible, presents a promising solution to meet the world's growing energy needs while reducing reliance on fossil fuels. However, the intermittent and variable nature of solar irradiance poses challenges for harnessing and utilizing solar power effectively.

In this context, the Buck-Boost converter emerges as a vital component in solar energy systems, offering versatility and efficiency in converting solar energy into a usable electrical supply. Unlike traditional voltage regulators, the Buck-Boost converter can step up or step down the input voltage to maintain a stable output voltage, making it well-suited for applications where input voltages fluctuate widely, such as solar photovoltaic (PV) systems.

The primary objective of this project is to design and simulate a Buck-Boost converter optimized for solar energy conversion. By regulating output voltage levels efficiently across a wide range of input voltages, the Buck-Boost converter maximizes energy extraction from solar panels, thereby enhancing the overall efficiency and performance of solar energy systems. Additionally, the project aims to analyze key parameters such as efficiency, voltage ripple, and control mechanisms to ensure the converter's effectiveness under varying solar irradiance conditions.

This introduction sets the stage for a comprehensive discussion on the objectives, methodology, and outcomes of the current project. By addressing the critical need for efficient energy conversion technologies in solar applications and highlighting the role of Buck-Boost converters in addressing these challenges, this project aims to contribute to the advancement of renewable energy technologies and the realization of a sustainable energy future.

Literature Review

Buck-Boost converters have garnered significant attention in the field of renewable energy systems due to their versatile voltage regulation capabilities and high efficiency. Previous research has explored various designs and applications of Buck-Boost converters, aiming to address the challenges associated with energy conversion in renewable energy systems, particularly in solar applications.

Efficient Energy Conversion: One of the primary focuses of previous research has been on improving the efficiency of Buck-Boost converters in solar energy systems. Studies have investigated different control strategies and circuit topologies to enhance the converter's efficiency, thereby maximizing energy extraction from solar panels. Techniques such as maximum power point tracking (MPPT) algorithms and soft-switching techniques have been explored to optimize energy conversion efficiency under varying solar irradiance conditions.

Voltage Ripple Reduction: Another area of research in Buck-Boost converters is the reduction of output voltage ripple. High voltage ripple can adversely affect the performance of downstream electronic devices and may lead to instability in the power supply. Research efforts have focused on minimizing voltage ripple through innovative circuit designs, component selection, and control algorithms. Techniques such as interleaved converters and active ripple cancellation methods have been proposed to mitigate voltage ripple and improve the overall stability of the power supply.

Control Mechanisms and Stability: Control mechanisms play a crucial role in ensuring the stability and reliability of Buck-Boost converters in renewable energy systems. Previous studies have investigated various control strategies, including pulse width modulation (PWM), frequency modulation (FM), and hybrid control schemes, to regulate output voltage levels and maintain system stability. Advanced control techniques such as predictive control and adaptive control have also been explored to improve the dynamic response and transient performance of Buck-Boost converters under changing operating conditions.

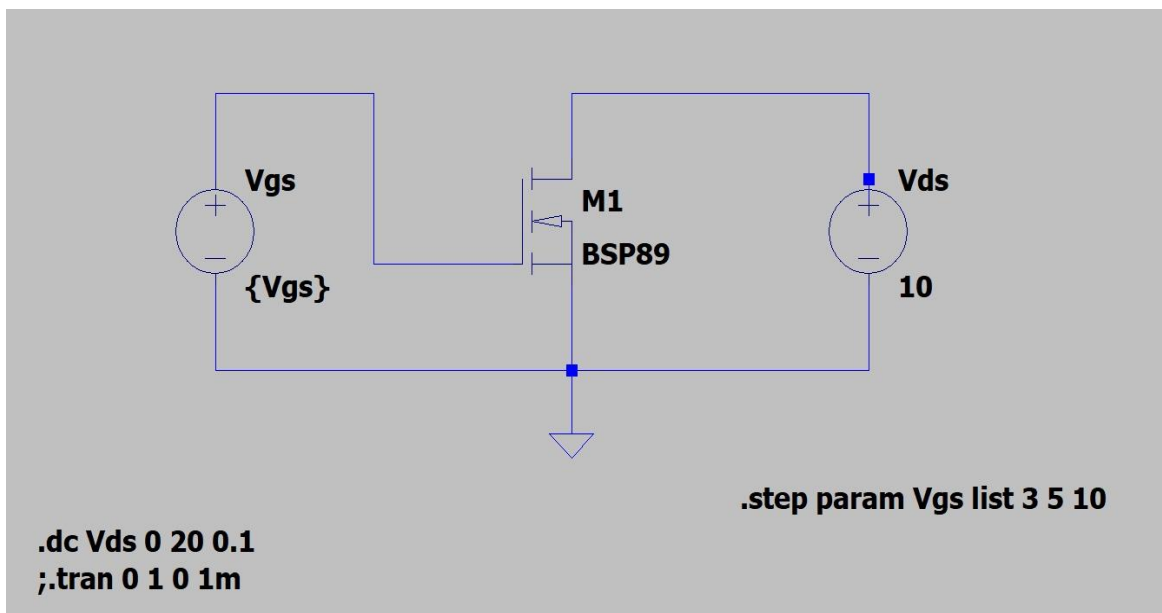
Integration with Renewable Energy Systems: Buck-Boost converters are widely used in renewable energy systems beyond solar applications. Research has explored their integration with other renewable energy sources such as wind, hydro, and fuel cells to create hybrid energy systems capable of efficiently harnessing multiple energy sources. Studies have investigated the optimal sizing and operation of Buck-Boost converters in hybrid energy systems to maximize overall system performance and reliability.

MOSFET Characteristics Analysis

OBJECTIVE

Efficient simulation of a Buck-Boost converter involves designing a circuit with a 24 VDC input, stepping it up to 30 VDC, and stepping it down to 9 VDC and 12 VDC through design and parameter using and simulation tools like MATLAB, P spice.

This report presents the findings from a simulation study of a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) to analyze its behavior under different gate-source voltages (V_{gs}). The study aims to illustrate the variations in drain current (I_d) with changes in drain-source voltage (V_{ds}) across three distinct V_{gs} values.



Circuit Configuration

The MOSFET model BSP89 was configured in a common source setup. A voltage source was connected to the gate terminal to provide varying V_{gs} , and another voltage source was applied across the drain-source terminals to vary V_{ds} . The simulation schematic is provided as part of the supplementary materials.

Simulation Parameters

Two key simulation commands were used:

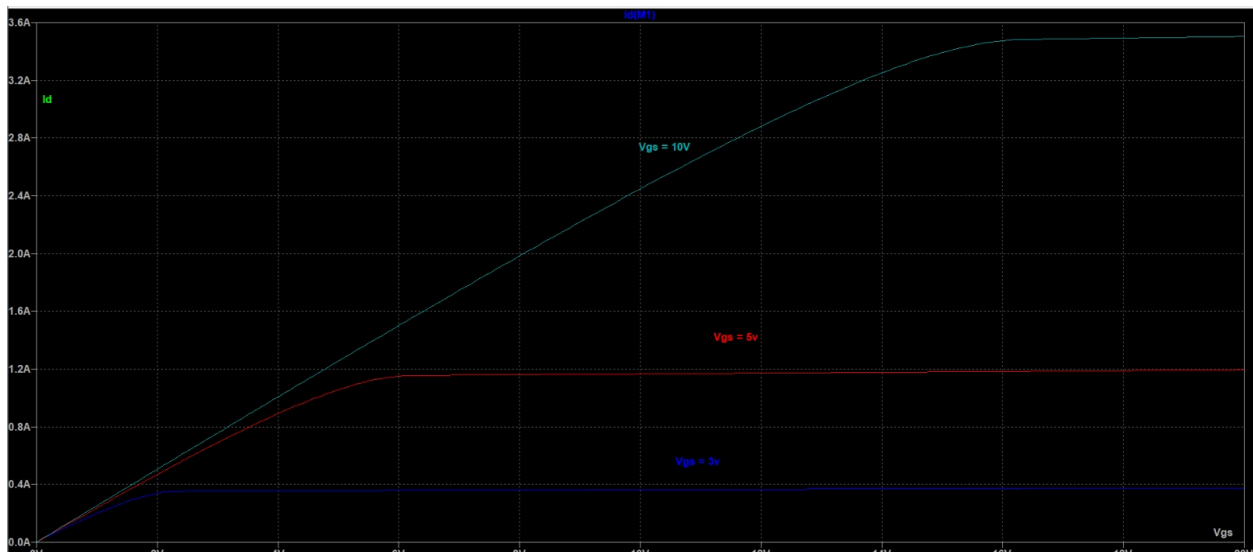
- A DC sweep command `.dc Vds 0 20 0.1`, which varied V_{ds} from 0 to 20V in 0.1V increments.
- A parametric sweep command `.step param Vgs list 3 5 10`, which varied V_{gs} in discrete steps of 3V, 5V, and 10V.

The transient analysis command in the schematic appears to be for illustrative purposes and was not relevant to the DC sweep results presented.

I_D vs. V_{gs} Characteristics:

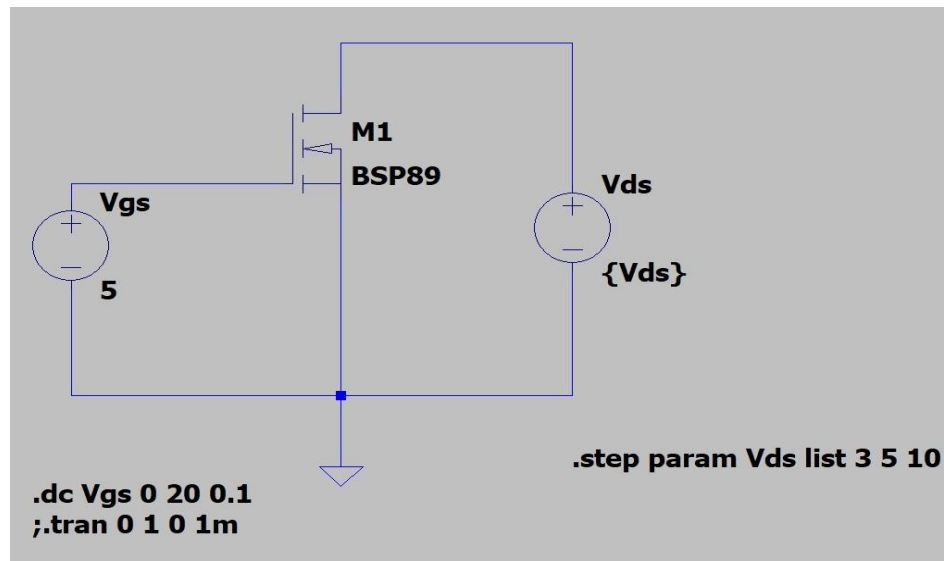
The plot of I_D against V_{ds} for different V_{gs} values demonstrates three operational regions for the MOSFET:

- For ($V_{gs} = 3V$) The plot reveals a modest increase in I_D with V_{ds} . The drain current is limited due to the lower V_{gs} , indicating that the MOSFET operates close to the cutoff threshold.
- For ($V_{gs} = 5V$) An elevated gate voltage leads to a more significant rise in I_D , showcasing an increased channel conductance and indicating a transition from the ohmic to the saturation region.
- For ($V_{gs} = 10V$) A steep rise in I_D is observed initially, followed by a leveling off. This indicates the MOSFET's quick entry into the saturation region where the current stabilizes despite further increments in V_{ds} .



MOSFET Drain Characteristics Study:

This section of the project report delineates the results obtained from the simulation study focused on the drain characteristics of a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET). The transistor model BSP89 was analyzed under various drain-source voltages (V_{ds}) while maintaining a constant gate-source voltage (V_{gs}).



Simulation Setup

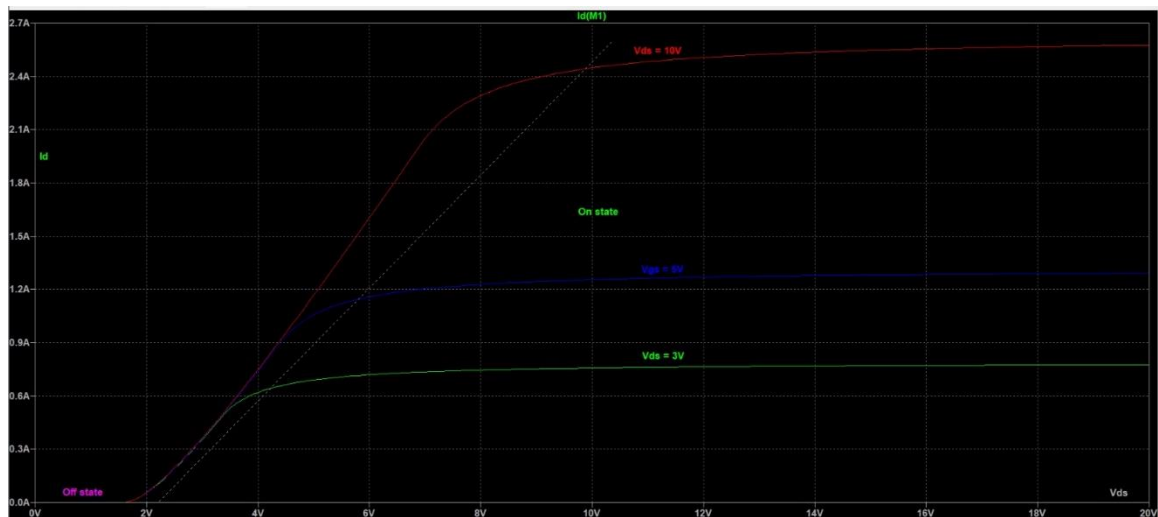
The study utilized a simulation schematic, where the MOSFET (M1) was set in a common-source configuration. The gate of M1 was supplied with a constant voltage (V_{gs}) of 5V, and a variable voltage source (V_{ds}) was applied across the drain-source terminals. The specifics of the simulation setup are depicted in the included schematic diagram.

Drain Current (I_d) Variation with Drain-Source Voltage (V_{ds})

The resulting plot captures I_d variation against V_{ds} at the set V_{gs} of 5V. The plot delineates three curves corresponding to V_{ds} values of 3V, 5V, and 10V, respectively, signifying different operating points of the MOSFET.

- $V_{ds} = 3V$ (green curve): Exhibits a gradual increase in I_d with V_{ds} , signifying the MOSFET's operation in the ohmic region where the current changes linearly with V_{ds} .

- $V_{ds} = 5V$ (blue curve): Demonstrates a transition behavior with a more pronounced increase in I_d compared to the 3V curve, suggesting the onset of the saturation region as V_{ds} increases.
- $V_{ds} = 10V$ (red curve): Shows the MOSFET firmly in the saturation region with a sharp increase in I_d initially, followed by a plateau, indicating the current is now relatively independent of V_{ds} .



The "On state" annotation on the plot implies that the MOSFET is fully active and conducting.

The study successfully charts the MOSFET's drain characteristics over a range of V_{ds} values at a constant gate voltage. It provides a clear understanding of how I_d behaves as the MOSFET transitions from the ohmic to the saturation region, a critical insight for utilizing MOSFETs in various electronic applications such as switching and amplification.

Analysis of Duty cycle for different voltage configurations:

Here we endeavor to design a buck-boost converter that efficiently manages the conversion of a fixed input voltage to diverse output voltages of 12V, 9V, and 30V. To ensure practicality and effectiveness in a variety of applications, the converter is designed to maintain a voltage ripple below 5%. The design process includes the careful selection of a switching frequency, as well as the sizing of critical passive components such as the inductor and capacitor. Moreover, the selection of an appropriately rated MOSFET is vital for handling the power requirements of the converter across the specified output voltage ranges. The aim is to achieve a versatile power supply solution that can adapt to varying loads while ensuring minimal loss of power and stability in performance.

Design Equations and Assumptions

Given an input voltage (V_{gs}) of 24V and a switching frequency (f_s) of 25kHz, the time period (T_s) is calculated as follows:

$$T_s = 1/f_s = 1/25e3 = 40\mu s.$$

The critical inductance (L_c) and capacitance (C_c) are calculated based on the duty cycle (D), with assumed load resistance (R) of 10 ohms, using the following formulas:

$$L_c = (1 - D)^2 R / 2 f_s$$

$$C_c = D * V / R * dv * f_s$$

Duty Ratio Analysis

Three cases are analyzed based on the required output voltage (V_o):

Case 1: $V_o = 12\text{ V}$

The duty cycle (D) is found using the buck-boost conversion ratio, resulting in a D of 0.33. Consequently, the critical inductance and capacitance are computed to be:

$$L_c = 89.78\mu\text{H}$$

$$C_c = 26.64\mu\text{F}$$

Case 2: $V_o = 9\text{ V}$

With a similar process, D is calculated to be 0.272, leading to:

$$L_c = 105\mu\text{H}$$

$$C_c = 26.64\mu\text{F}$$

Case 3: $V_o = 30\text{ V}$

For this case, D is determined to be 0.55, yielding:

$$L_c = 40.5\mu\text{H}$$

$$C_c = 44\mu\text{F}$$

Simulation Configuration

For the purpose of simulation, average or least values of L_c and C_c from the cases above were selected to maintain a single configuration. The simulation employed an inductance of $83.34\mu\text{H}$ and a capacitance of $55.55\mu\text{F}$. A MOSFET (IRFZ104) and a Schottky diode were incorporated into the circuit design.

The simulation was executed to verify the theoretical calculations and to observe the converter's performance. The obtained values of inductance and capacitance provided a stable output for a wide range of voltages. The results indicated that a single configuration could be utilized for different output requirements without the need to reconfigure the circuit for each case.

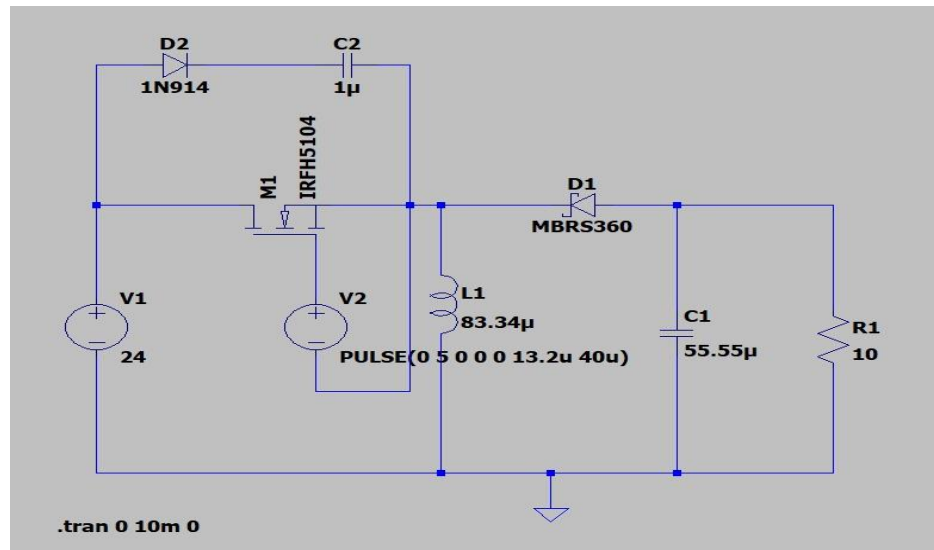
Further experimental validation is recommended to confirm the simulation results. In addition, a sensitivity analysis could be conducted to understand the impact of variations in component values on the converter's performance.

Attached to this report are detailed schematic diagrams and the simulation results that demonstrate the voltage regulation capabilities of the designed buck-boost converter circuit.

Simulation of a Buck-Boost Converter Circuit

The following report presents a comprehensive analysis of a simulated buck-boost converter circuit. The buck-boost converter is a type of DC-DC converter that can step up or down an input voltage to meet the required output voltage level. This type of converter is pivotal in applications where the power supply may vary, but a stable output voltage is necessary.

Circuit Configuration for 12V output voltage:



The simulated buck-boost converter comprised the following key components:

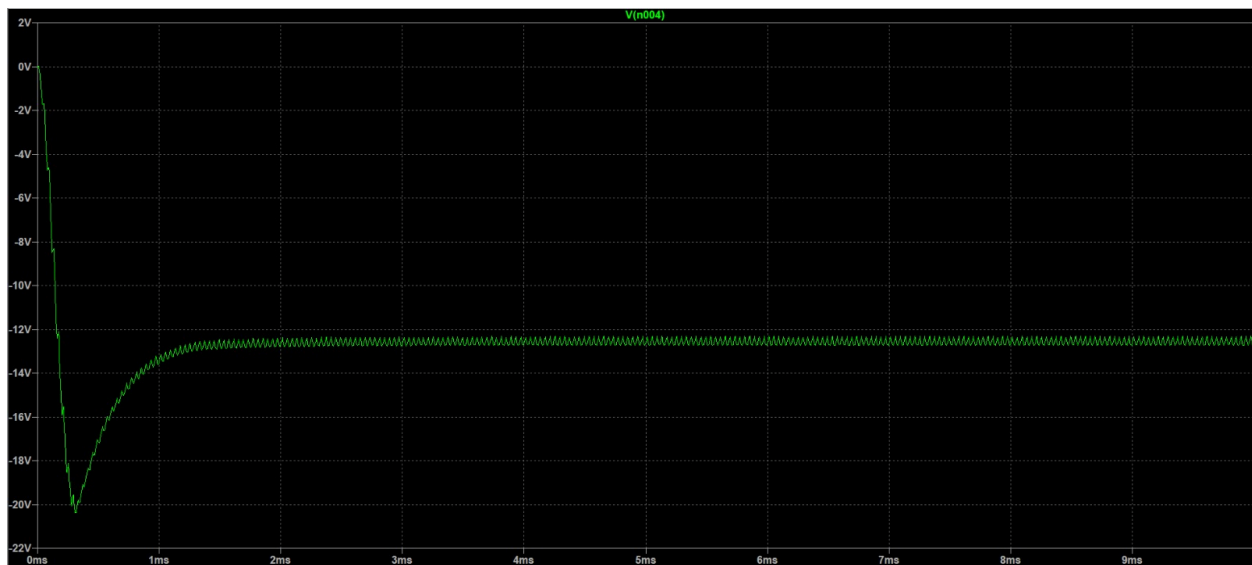
- V1: A 24V DC source, supplying power to the circuit.
- V2: A pulse signal with a pulse width modulated (PWM) control signal to drive the gate of the MOSFET (M1).
- M1 (IRFH1504): A MOSFET used as the switch to regulate the energy transferred to the inductor (L1).
- D1 (MBRS360): A Schottky diode to allow current flow during the non-conductive state of M1.
- D2 (1N914): A diode placed in parallel with V1 to prevent reverse current flow.
- L1: An inductor with a value of 83.34μH to store energy when M1 is conducting and release it when M1 is off.
- C1: A capacitor with a value of 55.55μH to smooth the output voltage.
- C2: A bypass capacitor of 1μF to stabilize the input voltage and reduce noise.
- R1: A 10-ohm resistor, representing the load.
- D : Duty ratio = 0.33
- T_s : Total time period = 40us

The operation mode of the converter is controlled by V2, with parameters defining the PWM signal's characteristics: a pulse of 0V to 20V with no rise, hold, or fall time, signifying a continuous conduction mode.

Simulation Procedure

The transient analysis was conducted over a 10ms duration to observe the output voltage waveform across the load, R1. The simulation results offer insights into the converter's efficiency and the stability of the output voltage.

Results and Observations



The simulation yielded a voltage waveform across R1. The plot shows a rapid initial drop in voltage as the circuit commences operation, followed by the waveform stabilizing into a steady-state pattern with periodic ripples. The voltage stabilizes at approximately -12V, which is indicative of the buck-boost operation where the output can be inverted and is lower than the input.

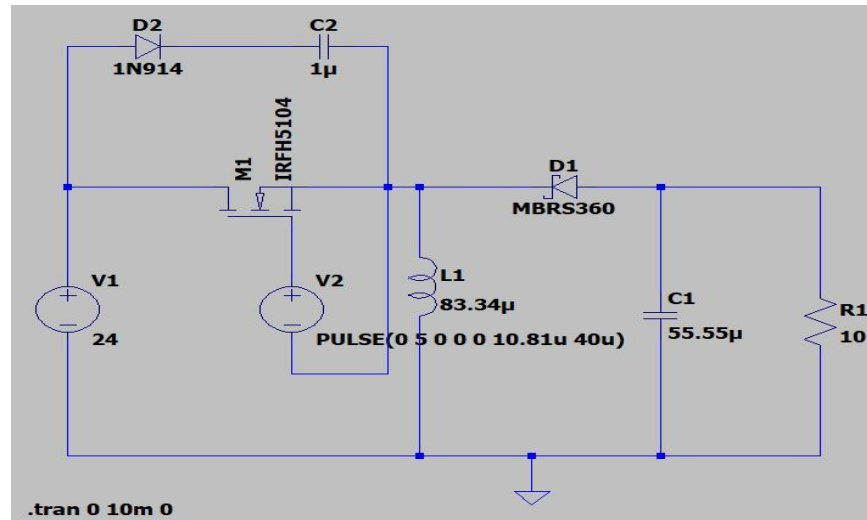
The observed initial dip in the voltage waveform corresponds to the inductive kickback when the MOSFET first switches on. Subsequently, the voltage ripple stabilizes, which is characteristic of a converter reaching its steady-state operation. The negative voltage indicates that the output is inverted relative to the input, a unique feature of the buck-boost topology.

The ripple magnitude, which is an important factor for the performance of sensitive electronic equipment, appears to be within an acceptable range for many applications. However, this would need to be confirmed with real-world measurements.

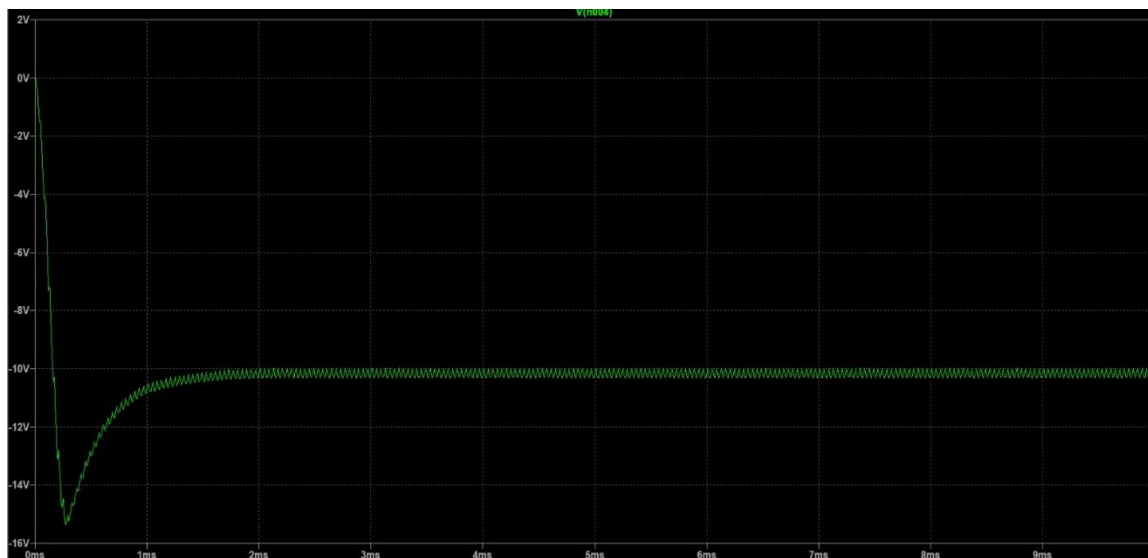
The simulated buck-boost converter has demonstrated the expected theoretical operation, with the ability to produce an inverted output voltage relative to the input. The results provide crucial data

that can be used to fine-tune the circuit for practical applications where precise voltage regulation and inversion are required.

Circuit Configuration for 9V output voltage:

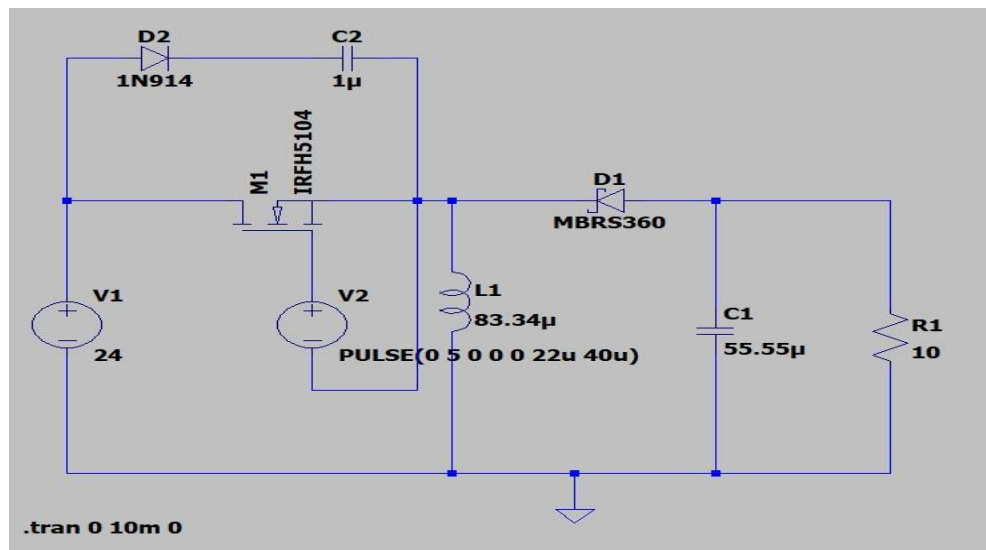


Results and Observation of case 2

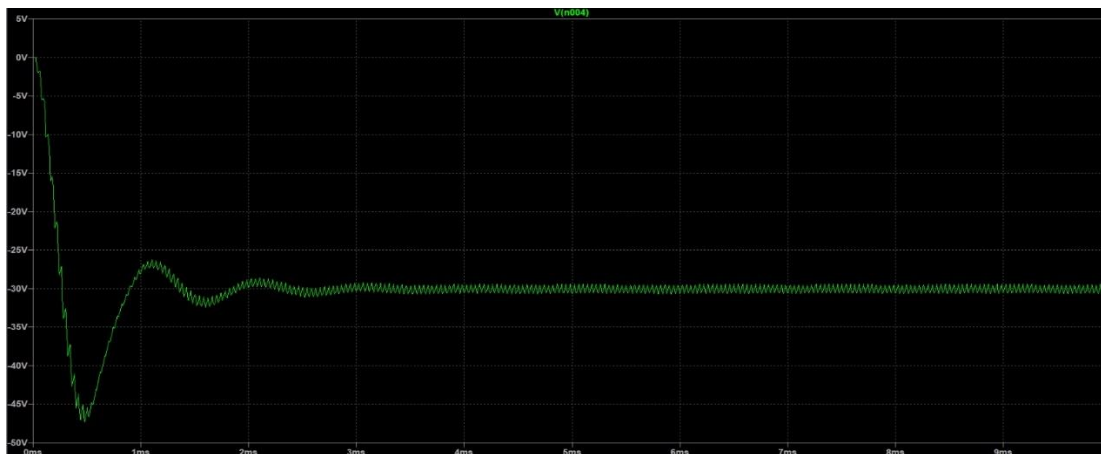


The simulation output plotted the voltage across the load (V(R1)) over time. The observed waveform showed an initial voltage spike, followed by a settling into a steady-state ripple. The average output voltage stabilized below the input voltage, characteristic of a buck-boost converter in buck mode, with the observed ripple due to the inductor and capacitor's energy exchange.

Circuit Configuration for 30V output voltage:

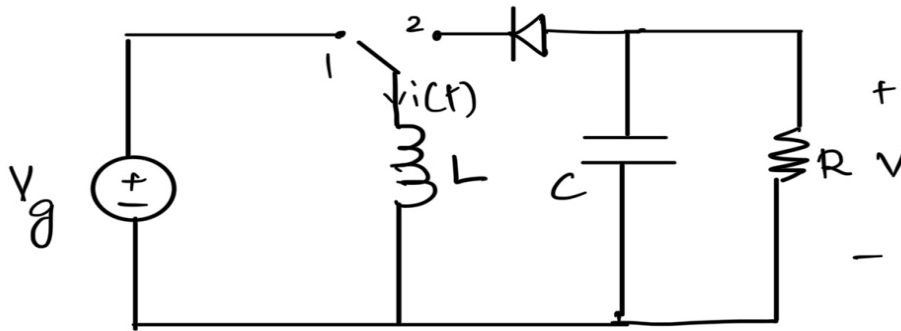


Results and Observation



The buck-boost converter operation was confirmed by the simulation results, demonstrating the expected output behavior. The voltage ripple magnitude is an important performance metric, as excessive ripple can affect the performance of downstream electronic devices. The simulation suggests that the chosen inductor and capacitor values result in a manageable ripple, although physical circuit testing would be necessary to validate this further.

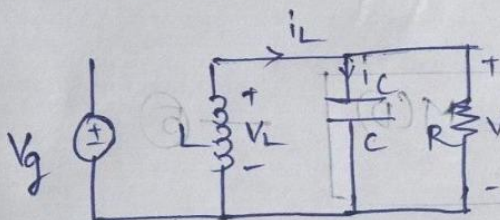
Analysis and MATLAB Visualization of Converter Ratio and Efficiency in Buck-Boost Converters



It outlines a step-by-step analytical approach to understanding the operational principles of a buck-boost converter, including the derivation of the converter ratio and its efficiency. This involves studying the converter in two distinct switch positions that define its operational states.

The below pictures describes about the analytical derivation to find the efficiency(η) and the converter ratio(M) of the converter.

Switch in position 1



$$V_L(t) = V$$

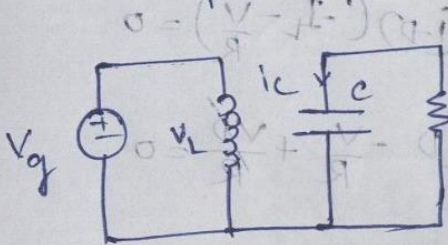
$$\frac{d}{dt} i_C(t) = -i_L - \frac{V}{R}$$

By small ripple approximation

$$V_L(t) \approx V \quad i_L \approx I$$

$$i_C = -I - \frac{V}{R} \quad V_L = V \quad (2)$$

Switch in position 2



$$V_L(t) = V_g$$

$$i_C = -\frac{V}{R}$$

By small ripple approximation

$$V_L \approx V_g \quad i_L \approx I$$

$$V_L = V_g \quad i_C = -\frac{V}{R} \quad (4)$$

The total area under inductor voltage and total area under capacitor current is zero when converter operates in steady state

$$\langle V_L \rangle = \frac{1}{T_s} \int_0^{T_s} V_L(t) dt = 0$$

$$\Rightarrow V_g D + V(1-D) = 0$$

$$V_g D - V_D + V = 0$$

$$(V_g - V)D + V = 0$$

$$(V_g - V)D = -V$$

$$V_g D = -V(1-D)$$

$$\boxed{\frac{V}{V_g} = \frac{-D}{1-D} = M(D)} \quad \text{--- (5)}$$

M(D) Converter Ratio for buck-boost converter

from (2), (4)

avg. capacitor current in steady state is zero

$$\langle i_C(t) \rangle = \frac{1}{T_s} \int_0^{T_s} i_C(t) dt = 0$$

$$\Rightarrow D(-V/R) + (1-D)(-I_L - V/R) = 0$$

$$-\frac{VD}{R} - I_L(1-D) - \frac{V(1-D)}{R} = 0$$

$$-\frac{V}{R} = I_L(1-D)$$

$$\text{Let } \frac{V}{R} = i_o$$

$$-i_o = I_L(1-D)$$

$$\boxed{\frac{I}{I_o} = \frac{1}{1-D}} \quad \text{--- (6)}$$

Assume no switching losses and no losses
ie all elements are ideal.
We find Efficiency for the converter

$$\text{Efficiency } (\eta) = \frac{P_{out}}{P_{in}}$$

$$P_{out} = V_{out} I_{out} = V_o \cdot I_o$$

$$P_{in} = V_{in} I_{in} = V_g \cdot I$$

$$\eta = \frac{V_o I_o}{V_g I}$$

$$= \frac{V_g \cdot \frac{D}{1-D} \cdot I_L (1-D)}{V_g \left(\frac{V_o}{V_g} \right) I}$$

$$= \frac{V_g \cdot D \cdot I_L}{V_g I}$$

$$\boxed{\eta = D} \quad \text{--- (7)}$$

From the above relation we can confirm that the efficiency of converter directly dependent on duty ratio of the converter

The buck-boost converter consists of an input voltage source (V_g), a switch, an inductor (L), a diode, a capacitor (C), and a load resistor (R). The switch toggles between two positions, changing the flow path of the current and alternately charging and discharging the inductor.

Switch at Position 1

With the switch in position 1, the inductor charges directly from the input voltage source. During this phase, the inductor current increases linearly with time, given by the equation derived from the loop analysis:

$$V_L = V$$

Where V_L is the voltage across the inductor, V_g is the input voltage, and V is the voltage across the load resistor.

Switch at Position 2

When the switch is in position 2, the inductor discharges through the load and the capacitor. The discharge path causes the inductor current to decrease. The voltage across the inductor in this state is derived using Kirchhoff's voltage law:

$$V_L = V_g$$

The equations also consider the effect of the capacitor by integrating the current over time, leading to the following relationship:

$$V_g D + V (1 - D) = 0$$

Derivation of Converter Ratio $M(D)$

The converter ratio $M(D)$ is defined as the ratio of the output voltage (V_o) to the input voltage (V_g). By analyzing the voltages during both the charging and discharging phases and assuming a steady-state operation where the average capacitor current is zero, the converter ratio is expressed as:

$$M(D) = V/V_g = D/(1-D)$$

This equation captures the dependency of the output voltage on the duty cycle D , which is the ratio of the switch conduction time to the total period.

Efficiency

The efficiency (η) of the converter is defined as the ratio of output power (P_{out}) to input power (P_{in}). Under the assumption of no switching losses and ideal circuit components, efficiency is simplified to:

$$\eta = P_{out}/P_{in}$$

By calculating P_{out} and P_{in} , efficiency can be found out as

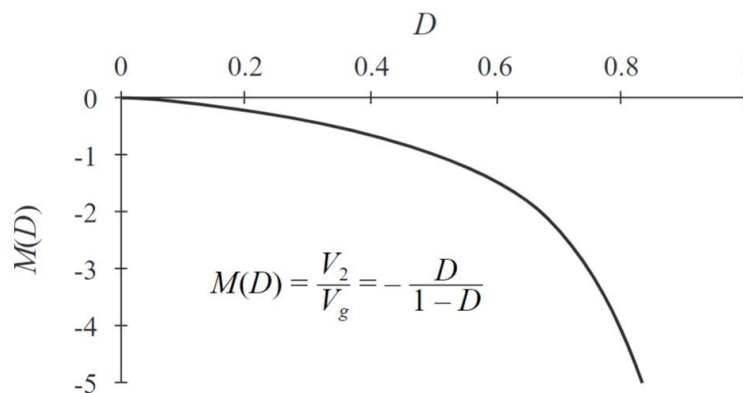
$$\eta = D$$

which is indicating that efficiency is directly proportional to the duty cycle of the converter.

MATLAB Visualization of Converter Ratio $M(D)$ for a Buck-Boost Converter

Converter Ratio Plot Analysis

Continuing from the analytical derivations previously discussed, MATLAB was used to visualize the converter ratio $M(D)$ for the buck-boost converter as a function of the duty cycle D . The converter ratio is graphed in the plot with the x-axis representing the duty cycle, ranging from 0 to 1, and the y-axis representing the converter ratio $M(D)$.



The MATLAB plot exhibits the relationship between $M(D)$ and D as per the derived formula:

$$M(D) = V/V_g = D/(1-D)$$

The curve shows a nonlinear relationship between the duty cycle and the converter ratio. As D approaches 1, the magnitude of $M(D)$ tends to increase steeply, indicating that small changes in the duty cycle near full conduction can cause large changes in the output voltage relative to the input voltage. The negative sign indicates that the output voltage is inverted with respect to the input voltage, which is a characteristic of the buck-boost converter.

The curve begins at $M(D) = 0$ when $D = 0$, signifying no output voltage as expected when the switch is always off. As the duty cycle increases, the output voltage becomes more negative with respect to the input, which corresponds to the buck-boost converter's ability to invert the output.

The plot validates the theoretical expectation that the output voltage of a buck-boost converter can be varied continuously by adjusting the duty cycle from zero to near unity. However, the practical limitations of the circuit components and the non-idealities such as switch and diode forward voltage drops are not accounted for in this idealized plot.